

SE implementation document

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Development Environment Setup and Implementation Guide

Student Mental Health Support System

Overview

This document outlines the setup and implementation for the full-stack development project designed for the **Student Mental Health Support System**, which tracks student mental health, counseling sessions, and wellness programs. The technologies used in the project are:

- **Backend:** Node.js with Express.js
- **Frontend:** Vite React with React Router DOM
- **Database:** MongoDB with Mongoose
- **Version Control:** GitHub
- **Authentication:** JSON Web Token (JWT) with bcrypt
- **Deployment:** Heroku (for basic deployment)

1. Backend Setup: Node.js with Express.js

The backend of the **Student Mental Health Support System** is built using Node.js and Express.js to manage server-side operations such as user authentication, managing counseling sessions, handling student data, and wellness resource tracking. This provides scalability and flexibility in handling multiple concurrent user requests securely.

API Design for the Mental Health System

Design RESTful APIs to handle the following requests:

- **User Authentication:** Use JWT to secure login and registration processes. Bcrypt is used to encrypt passwords before storing them in the database.
- **Counseling Session Management:** APIs for students to book, update, or cancel counseling appointments, and for counselors to view their schedules and manage sessions.
- **Mental Health Resources:** APIs to manage articles, guides, and videos that help students address specific wellness concerns.
- **Student Wellness Surveys:** APIs to manage wellness survey submissions, and store results for analysis and personalized recommendations.
- **Role-based Authorization:** Allow students, counselors, and administrators to access relevant data based on their roles using RBAC (Role-Based Access Control).

Key Packages/Dependencies

- **Express.js:** Lightweight web framework for Node.js.
- **Mongoose:** MongoDB object modeling for Node.js.
- **JWT & Bcrypt:** For authentication and password encryption.

2. Frontend Setup: Vite React with React Router DOM

The frontend of the system is built using **Vite React**, which offers fast development with hot reloading and smooth performance. **React Router DOM** is used to manage navigation between different sections of the app, ensuring a dynamic, single-page application experience.

Key Features

- **Student Dashboard:** Provides an overview of booked counseling sessions, wellness recommendations, and access to mental health resources.
- **Counselor Dashboard:** Allows counselors to view and manage their schedules, with session details for each student.
- **Admin Panel:** Allows the system administrator to manage users, counselors, and mental health resources.
- **Responsive Design:** Ensures the system is accessible on both desktop and mobile devices.

Key Packages/Dependencies

- **Axios:** For making HTTP requests to the backend APIs.
 - **React Router DOM:** For implementing client-side routing.
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3. Database Setup: MongoDB with Mongoose

MongoDB is used to store data related to students, counselors, counseling sessions, and wellness resources. **Mongoose** provides an elegant schema-based solution to model this data in MongoDB.

Key Data Entities

- **Users:** Includes student profiles (e.g., name, email, enrollment number), counselor profiles (e.g., specialization, availability), and admin roles.

- **Counseling Sessions:** Includes details of booked sessions such as session ID, student ID, counselor ID, date, and status (booked, rescheduled, canceled).
- **Wellness Surveys:** Stores student-submitted wellness data and calculates results to suggest resources or sessions.
- **Mental Health Resources:** Articles, videos, and guides categorized by topic (e.g., stress management, anxiety) for students.

Relationships

- A student can book multiple counseling sessions with a counselor.
 - Wellness surveys are linked to the student's profile, with recommendations updated based on survey results.
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4. Version Control: GitHub

GitHub is used for version control to enable collaboration, branch management, and tracking of code changes. It helps manage feature development (authentication, session management, wellness resources) and bug fixes through separate branches.

Key GitHub Features

- **Feature Branches:** Separate branches for different functionalities, such as user authentication, session booking, and resource management.
 - **Pull Requests:** Collaborators can request reviews before merging changes into the main branch, ensuring code quality and reducing the chances of introducing bugs.
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5. Authentication: JSON Web Token (JWT) and Bcrypt

Authentication is managed using **JWT** to create secure tokens that verify user identities. **Bcrypt** hashes and salts passwords to ensure secure storage in the database.

Authentication Flow

- Users (students, counselors, admins) register by creating an account, after which passwords are securely hashed using bcrypt.
 - Upon successful login, a JWT token is generated and stored client-side to handle future authentication.
 - The token expires after a specified period and can be refreshed securely to continue sessions.
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6. Deployment: Heroku

Heroku is used for deployment due to its simplicity and free-tier hosting options for both backend and frontend. The deployment is separated into two applications:

1. **Backend Deployment:** The Node.js server is deployed on Heroku with environment variables like MongoDB URI and JWT secret set up correctly.
2. **Frontend Deployment:** The Vite React application is deployed on Heroku, connected to the backend via the REST APIs.

Deployment Steps

- Push the code to a GitHub repository.
- Connect the repository to Heroku for automatic deployment.
- Configure environment variables on Heroku for backend (e.g., MongoDB URI, JWT secret).

- Ensure that both backend and frontend are correctly connected through API URLs in production.

This guide provides a comprehensive approach to developing and deploying the **Student Mental Health Support System** using modern web technologies.